

Inverse PINN Results for Young's Modulus Estimation

Lamb-wave simulation data | Pristine 1 mm plate | 120 kHz | 15 dB noise

Training Set-up (PINN)

- Training dataset: 30 known Young's modulus cases
- Training modulus range: 1–100 GPa
- Wave data: Pristine plate, 120 kHz, 15 dB noise
- Data type used: Velocity Training steps: 5,000
- Final PDE loss: 7.065×10^{-14}

Unknown Test Results

Inverse PINN Results on Unknown Moduli

Unknown E	Predicted E	Absolute Error	Error Percentage	Inverse steps
50.00 GPa	51.0250 GPa	1.0250 GPa	2.05%	10,000
91.00 GPa	51.0241 GPa	39.9759 GPa	43.93%	10,000

The PDE value shown in Slide 2 comes from training. The current inverse code reported data loss, not a separate PDE value for each unseen test.

Observations

- The model estimated the unseen 50 GPa case accurately, with 2.05% error.
- The model did not estimate the unseen 91 GPa case accurately, with 43.93% error.
- Both inverse runs ended near 51 GPa.
- The inverse optimization starts near the midpoint of the 1–100 GPa training range, approximately 50.5 GPa.
- The result suggests that the current inverse setup is more sensitive near the middle of the training range than near the high end.
- Next step: test multiple inverse starting guesses and retrain in a clean folder for a reproducible experiment.